📊 *Data Analyst Candidate Selection Tool using Python, Pandas, and Matplotlib*
📄 *Project Report*
👨‍💻 *Prepared by:* Saurav Prakash Shejwal
💻 *Platform:* Google Colab
🔗 *Programming Language:* Python
📚 *Libraries Used:* Pandas, Matplotlib, OpenPyXL
🎯 *Project Type:* Data Analytics Project

📌 *1. Project Overview*
The Data Analyst Candidate Selection Tool is a Python based data analytics project designed to analyze candidate information stored in an Excel file. The project filters candidates with Data Analysis skills, exports selected candidates to a new Excel file, and visualizes candidate skill distribution.
This project simulates a real world HR analytics workflow used by organizations to identify suitable candidates.

📂 *2. File Preparation and Renaming*
📌 *Purpose:*
Before starting the project, the Excel dataset must be renamed properly to avoid errors.
✅ *Step performed:*
Rename the Excel file to:
candidates.xlsx
⚠️ *Important Guidelines:*
• Avoid spaces in file names
• Avoid brackets or special characters
• Use simple and clean file names
Final file name used in project:
candidates.xlsx

📤 *3. Upload Dataset in Google Colab*
📌 *Purpose:*
This step uploads the dataset into the Colab environment.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files top_data_a…304 (1).xlsx
top_data_analyst_candidates_20231217_003304 (1).xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 5128 bytes, last modified: 2/24/2026 - 100% done
```
Saving top_data_analyst_candidates_20231217_003304 (1).xlsx to top_data_analyst_candidates_20231217_003304 (1).xlsx
```

📖 *Explanation:*

This command opens a file upload dialog and allows the user to upload the Excel dataset.

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4. Import Required Libraries

Purpose:

This step imports necessary libraries for data analysis and visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
from datetime import datetime
```

Explanation:

- pandas → Used for reading and analyzing Excel data
- matplotlib → Used for data visualization
- datetime → Used for generating timestamp based filenames

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5. Reading the Excel Dataset

Purpose:

This step loads the Excel file into a Pandas DataFrame.

```
df = pd.read_excel("candidates.xlsx")
df.head()
```

	Name	Mobile Number	Skill	Age
0	Munna	9871811905	Data Analysis - Expert	51
1	Anjali	9419197898	Data Analysis - Expert	47
2	Beena	9316053430	Data Analysis - Intermediate	57
3	Munna	9971335733	Data Analysis - Expert	34
4	Munna	9764766567	Data Analysis - Intermediate	38

Next steps: Generate code with df | New interactive sheet

Explanation:

- Reads Excel file
- Displays first five rows
- Helps understand dataset structure

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6. Cleaning Column Names

Purpose:

Removes hidden spaces from column names to prevent errors.

```
df.columns = df.columns.str.strip()
print(df.columns)
```

```
Index(['Name', 'Mobile Number', 'Skill', 'Age'], dtype='object')
```

📖 *Explanation:*

Ensures column names are clean and usable.

🎯 *7. Filtering Data Analyst Candidates*

📌 *Purpose:*

Filters candidates who have Data Analysis skills.

```
data_analyst_candidates = df [
    df['Skill'].str.contains('Data Analysis', na = False)
]

data_analyst_candidates = data_analyst_candidates [
    ['Name','Mobile Number', 'Skill', 'Age']
]

data_analyst_candidates = data_analyst_candidates.head(5)

data_analyst_candidates
```

	Name	Mobile Number	Skill	Age
0	Munna	9871811905	Data Analysis - Expert	51
1	Anjali	9419197898	Data Analysis - Expert	47
2	Beena	9316053430	Data Analysis - Intermediate	57
3	Munna	9971335733	Data Analysis - Expert	34
4	Munna	9764766567	Data Analysis - Intermediate	38

Next steps: Generate code with `data_analyst_candidates` New interactive sheet

📖 *Explanation:*

- Filters candidates with Data Analysis skill
- Selects relevant columns
- Displays top 5 candidates

💾 *8. Exporting Filtered Candidates to Excel*

📌 *Purpose:*

Saves filtered candidates into a new Excel file.

```
timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")

filename = f"top_data_analyst_candidates_{timestamp}.xlsx"

data_analyst_candidates.to_excel(filename, index=False)

print("File saved as:", filename)
```

```
File saved as: top_data_analyst_candidates_20260224_061819.xlsx
```

📖 *Explanation:*
Creates automated Excel report with timestamp.

⬇️ *9. Downloading Result File*
📌 *Purpose:*
Downloads generated Excel report.

```
files.download(filename)
```

📖 *Explanation:*
Allows saving the generated report locally.

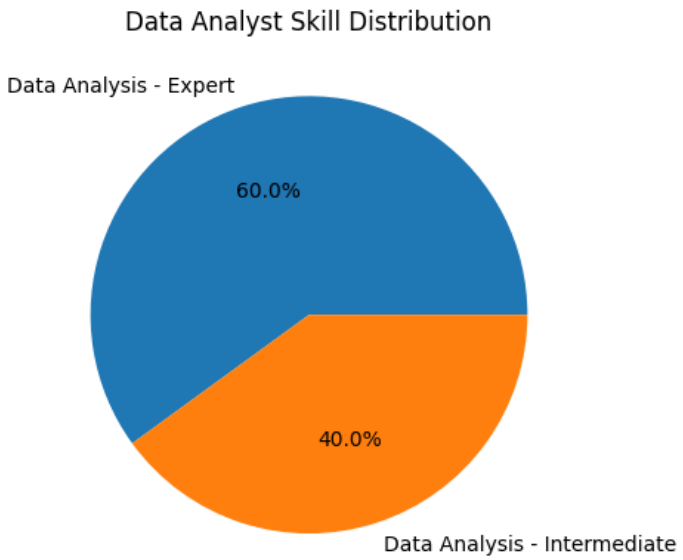
📊 *10. Visualizing Skill Distribution using Pie Chart*
📌 *Purpose:*
Shows distribution of candidate skills.

```
data_analyst_candidates['Skill'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Data Analyst Skill Distribution")

plt.ylabel("")

plt.show()
```



📖 *Explanation:*
Helps visualize candidate skill proportions.

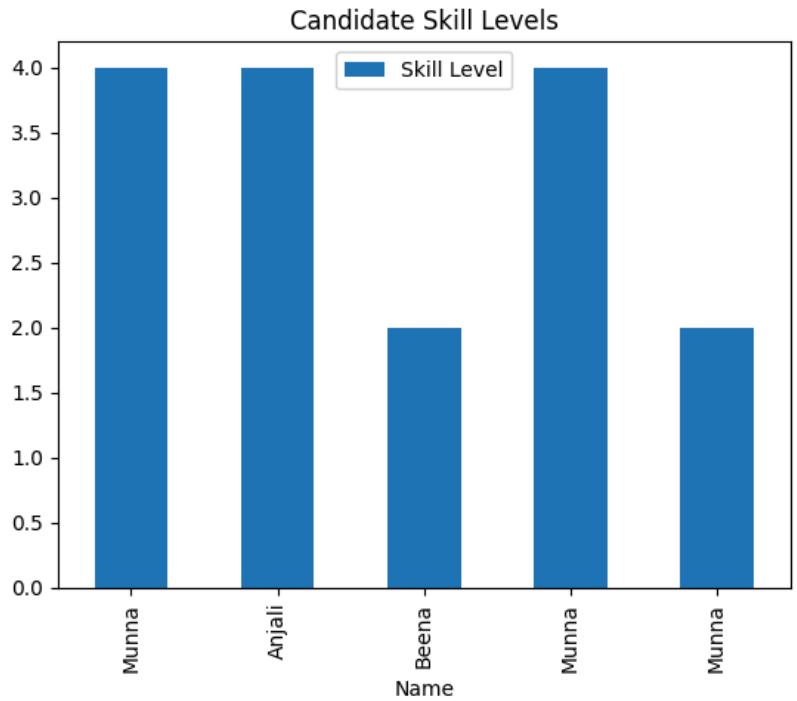
📈 *11. Visualizing Skill Levels using Bar Chart*
📌 *Purpose:*
Compares candidate skill levels.

```
skill_levels = {
    'Beginner': 1,
    'Intermediate': 2,
    'Advanced': 3,
    'Expert': 4
}

data_analyst_candidates['Skill Level'] = data_analyst_candidates[
    'Skill'
].apply(lambda x: skill_levels.get(x.split(' - ')[1], 0))

data_analyst_candidates.plot(
    x='Name',
    y='Skill Level',
    kind='bar',
    title='Candidate Skill Levels'
)

plt.show()
```



📖 *Explanation:*

Converts skill levels to numeric values and visualizes comparison.

✅ *12. Results*

The project successfully performed the following tasks:

- Loaded dataset from Excel
- Cleaned dataset
- Filtered Data Analyst candidates
- Exported filtered data
- Generated visualizations

🎯 *13. Conclusion*

This project demonstrates a complete real world data analytics workflow using Python.

Skills demonstrated:

- Data analysis using Pandas
- Excel file handling
- Data filtering

- Data visualization
- Automated report generation

This project is highly relevant for Data Analyst roles.

🚀 *14. Future Scope*

Possible improvements include:

- Candidate scoring system
- Interactive dashboard
- Database integration
- Web application using Streamlit

📌 *15. Final Summary*

This project successfully demonstrates how Python can be used to analyze candidate data, generate reports, and visualize insights using industry standard tools.